

LIVER HEPATOCYTE

Linoleic Acid (18:2n-6)

FADS1/2/6 · ELOVL2/5/6
Tier 1: FADS1, FADS2, FADS6 $r=0.75-0.84$

Arachidonic Acid (20:4n-6)

ACSL4 → AA-CoA (Lands Cycle)
Tier 1: ACSL4 $r=0.746$

Membrane PL → PLA2G6 → FFA + AA
Tier 1: COX/PTGDS $r=0.808$ LOX/GPX CYP450

CBR2 → PGE2/PGD2/TXA2
PTGDS → PGD2 $r=0.927$

GPX3 → LTC4/LTC4
→ Leukotrienes $r=0.839$

CYP2E1/CYP4V2 → EETs
EETs/HETEs $r=0.772-0.852$

SKELETAL MUSCLE

Membrane GPCRs

EP4 (PTGER4) — PGE2 receptor

TP (TBXA2R) — TXA2 receptor

BLT (LTB4R) — LT receptor

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Second Messengers

cAMP-PKA / Ca²⁺-PKC

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Nuclear Receptors

PPARalpha (EET sensor)

PPARα

PORTAL / SYSTEMIC CIRCULATION

Eicosanoids as Endocrine Signals
(PGE2, PGD2, TXA2, LTs, EETs)

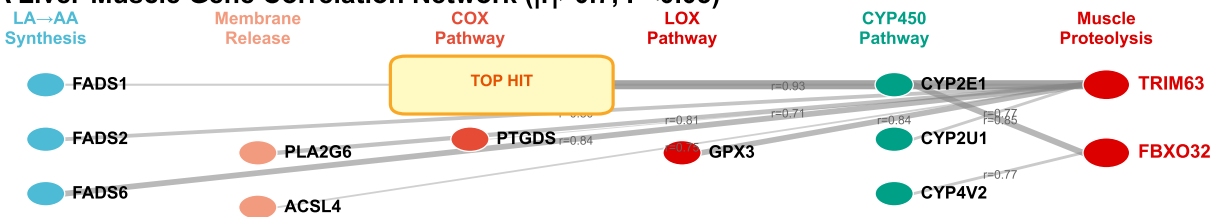
FOXO3 — Transcription Factor

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TRIM63/MuRF1 + FBXO32/Atrogin-1

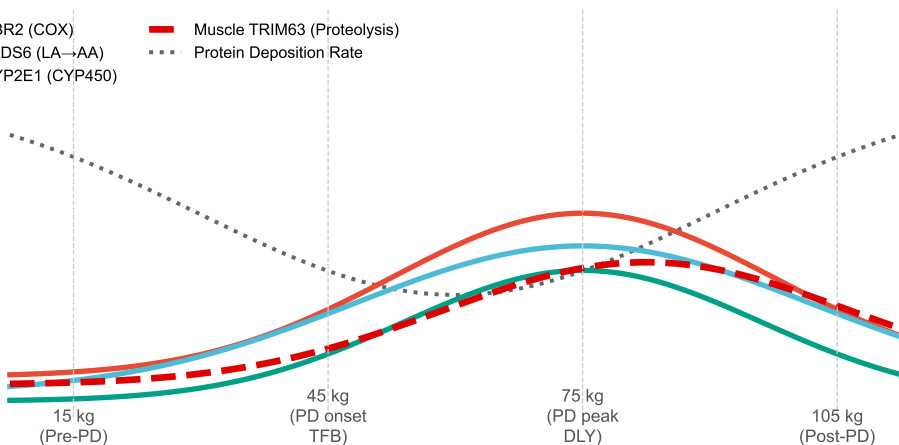
Ubiquitin-Proteasome → ↑ Protein Degradation

B. AA Liver-Muscle Gene Correlation Network ($|r|>0.7$, $P<0.05$)



C. Hypothetical Signal Cascade Model

— Liver CBR2 (COX) — Muscle TRIM63 (Proteolysis)
— Liver FADS6 (LA→AA) — Protein Deposition Rate
— Liver CYP2E1 (CYP450)



Proposed Mechanism:

- (1) Growth-stage-dependent upregulation of liver AA metabolic enzymes (FADS1/2/6, PLA2G6, CBR2, GPX3, CYP2E1)
- (2) Increased hepatic eicosanoid production (PGE2, PGD2, LTs, EETs) released into circulation
- (3) Activation of muscle GPCRs (EP4/TP/BLT) → cAMP-PKA / Ca²⁺-PKC cascades
- (4) PPARalpha activation by EETs → FOXO3 transcriptional program
- (5) FOXO3 drives TRIM63 (MuRF1) + FBXO32 (Atrogin-1) → ubiquitin-proteasome → protein degradation
- (6) This axis counterbalances anabolic insulin/IGF-1 signaling at the protein degradation node